Bibliometric and Temporal Analysis

Bibliometric and Temporal Analysis I

Descriptive statistics summarize the corpus along every available axis: counts by year, venue, and author; citation-count distributions; and author productivity. Temporal analysis fits the publication time series, reporting a compound annual growth rate of 5.48% across 2000–2026 (a span of 26 years), with a mean year-over-year growth rate of 7.8% and a doubling time of 9.2 years. The peak publication year is 2025 with 147 records.

Growth Metrics I

The compound annual growth rate (CAGR) is computed as:

$$\text{CAGR} = \left(\frac{N_{\text{end}}}{N_{\text{start}}}\right)^{1/(\text{year span})} - 1$$

where N_{start} is the publication count in the first year (2000) and N_{end} is the count in the last year (2026). The mean year-over-year growth rate $\bar{g}$ is the arithmetic mean of annual ratios. The doubling time is $t_d = \ln(2)/\ln(1 + \text{CAGR})$. These metrics are stored in `temporal_analysis.json` and injected into the manuscript at render time.

Subfield Classification I

Subfield classification assigns each record to one of 6 configurable buckets (Clinical Sleep, Cognition, Pharmacology, Psychiatry, Safety, and Neuroscience) by priority-aware keyword matching; the taxonomy is defined entirely in configuration (`project_config.subfield_keywords`). The largest bucket is **Clinical Sleep** at 63.0% of the classified corpus. A per-subfield temporal breakdown (`subfield_timeline.json`) tracks how each sub-area has grown over time, enabling identification of emerging or declining research threads.

Topic Modeling I

A TF-IDF term-weighting of titles and abstracts [@salton1988term] feeds non-negative matrix factorization (NMF) [@lee1999learning], implemented with scikit-learn [@pedregosa2011scikit]. NMF decomposes the document-term matrix $\mathbf{V} \approx \mathbf{W}\mathbf{H}$, where $\mathbf{W}$ is the document-topic matrix and $\mathbf{H}$ is the topic-term matrix. The factorization extracts 5 latent topics that cross-cut the keyword taxonomy. The random seed is fixed at 42 for reproducibility. The reporting follows established systematic-review practice [@page2021prisma], with every figure and statistic traceable to a committed artifact.